

Ashutosh Roy

AI/ML Researcher · NLP & Multimodal Systems · Published at IEEE RECCAP 2026 & IIT Delhi

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Summary

AI/ML researcher and engineer with **2 published papers** (RECCAP 2026, IIT Palakkad; co-authored with IIT Delhi). Hands-on experience building large-scale NLP, multimodal AI, RAG, and speech processing systems. Skilled in transformer fine-tuning, hybrid retrieval pipelines, and end-to-end ML system deployment. Proficient in **Python, C++**, PyTorch, TensorFlow, and LangChain. Open-source contributor with production projects serving real users. Seeking to contribute to Google's research agenda on real-world, large-scale AI problems.

Experience

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built an OCR-powered document understanding pipeline for archaeological datasets; benchmarked Tesseract, Marker, and Gemini OCR — selecting optimal strategy per document class to improve extraction accuracy.
- Designed hybrid retrieval (dense: FAISS/Pinecone + sparse: Apache Solr) for high-precision semantic + lexical document search over thousands of historical records.
- Deployed scalable **FastAPI** services for real-time LLM-based querying; integrated with frontend for sub-second retrieval responses.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi collaborators — submitted for peer review.
arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned transformer-based NLP models (Mistral/LLaMA variants) for healthcare applications — medical record summarization and symptom-driven conversational AI.
- Reduced patient query resolution time by **~40%** by deploying an AI chatbot for telemedicine with optimized intent understanding and response generation.
- Implemented GenAI-powered NLP workflows using LLMs and retrieval-augmented pipelines for intelligent healthcare automation and conversational AI systems.
- Built end-to-end ML pipelines: data preprocessing → model inference → API deployment for real-time healthcare applications.

Publications & Research

SARCH: Multimodal Search for Archaeological Archives | Co-authored with IIT Delhi (Prof. Maya Ramanath)

2025

- Proposed a multimodal retrieval system combining OCR (Surya), image classification, and hybrid search (FAISS + Apache Solr) for large-scale scanned archaeological archives.
- Benchmarked multiple OCR engines (Tesseract, Marker, Gemini OCR) across diverse historical document layouts; achieved significant gains in extraction accuracy.

Multitask Speech Emotion & Stress Detection (Wav2Vec2-BiLSTM) | Accepted — RECCAP 2026, IIT Palakkad (IEEE)

2026

- Proposed a self-supervised Wav2Vec2 + BiLSTM multitask architecture for simultaneous 5-class emotion classification and continuous stress regression; achieved **85.9% accuracy**, **0.856 F1**, RMSE **0.1268**.
- Evaluated across 3 corpora (RAVDESS, TESS, SAVEE) with unified preprocessing pipeline (VAD, silence removal, log-Mel spectrograms) and data augmentation.

Projects

Archaeological Full-Stack Search Engine | Python, Surya OCR, Apache Solr, FAISS, FastAPI, C++

Jan 2025 – Jun 2025

- Built an end-to-end search engine for scanned archaeological texts and maps at IIT Delhi; fine-tuned Surya OCR to convert multi-layout historical books into structured Markdown/JSON.
- Extracted semantic geographic and historical metadata from maps using NLP pipelines; indexed output with Apache Solr for sub-second keyword and semantic retrieval.
- Demo video available; system deployed with FastAPI backend serving real-time search queries.

Exam-Helper RAG System | Python, LLaMA 3, FAISS, Pinecone, LangChain, Streamlit

Dec 2024 – Apr 2025

- Built a production-ready LLM-powered contextual QA system on **LLaMA 3** with RAG; multimodal ingestion pipeline supports PDFs, images, and YouTube transcripts.
- Implemented dense retrieval via FAISS/Pinecone for sub-second similarity search, reducing hallucination and improving context precision over baseline.

Shopping Assistant Agent | Python, LangGraph, MCP, FastAPI

Sep 2024 – Dec 2024

- Engineered a multi-agent shopping assistant using **LangGraph** and MCP servers for intelligent product search and recommendation workflows with self-reflection and retry mechanisms.

Medical Chatbot (LLM Fine-Tuning) | Python, Mistral 7B, QLoRA, Tesseract OCR

Mar 2024 – Apr 2024

- Fine-tuned **Mistral 7B** via **QLoRA** (4-bit quantization) for domain-specific medical NLU; reduced compute cost vs. full fine-tuning while preserving task accuracy.
- Built OCR pipeline using Tesseract to extract and structure information from unstructured clinical reports for downstream LLM inference.

Open Source Contributions

- InterviewPrep — Open-Source AI/ML Interview Platform | Astro, JavaScript, HTML/CSS, SEO

May 2026 – Jun 2026

 - Developed and open-sourced a comprehensive interview prep platform with **700+ curated questions** spanning Python, ML, Deep Learning, NLP, LLMs, RAG, Computer Vision, SQL, and HR.
 - Features company-specific tracks (FAANG, OpenAI, NVIDIA, Indian IT), a 12-week structured roadmap, interactive flashcards, and role-specific mock interview simulations.
 - Fully responsive static site with light/dark mode, optimized schema markup, and **95+ Lighthouse performance score**; designed for large-scale community use.
- Skills-for-Agents — Composable Instruction Sets for Multi-Task AI Agents | Python, LLM Agents, SIP Protocol

May 2026

 - Built a production-ready ecosystem of modular, conflict-free instruction sets for multi-task LLM agents; each skill owns exactly one domain (voice, density, craft, process, or content).
 - Designed a custom **Skills Interoperability Protocol (SIP)** to completely prevent instruction conflicts during concurrent multi-agent execution, enabling scalable and reliable agent workflows.
- BulkyMail — Automated Bulk Email Campaign Tool | Python, Streamlit, SMTP/API

Dec 2024

 - Built a Streamlit web app for automated, personalized bulk email campaigns; enabled users to send **100+ personalized emails in minutes** with dynamic templating (name, role, company).
 - Contributed to successful internship conversions for multiple users by improving outreach response rates through personalized messaging at scale.

Technical Skills

AI/ML & Research: LLMs, GenAI, RAG, NLP, Transformer Fine-Tuning (QLoRA/LoRA), Self-Supervised Learning, Multimodal AI, Speech Processing, Computer Vision, Dense Retrieval, Foundation Models, Evaluation & Benchmarking

Languages: Python, C++, SQL, JavaScript

Libraries & Frameworks: PyTorch, Hugging Face Transformers, LangChain, TensorFlow, Scikit-learn, Pandas, NumPy, FastAPI, Streamlit, OpenCV

Databases & Search: FAISS, Pinecone, Apache Solr, MySQL

SDLC & Engineering: Git, Docker, Linux, REST API Design, MLOps, Model Deployment, Agile workflows, CI/CD pipelines

Tools: Jupyter Notebook, Google Colab, VS Code, OpenAI GPT, Claude, Power BI, MATLAB

Education

- Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh
- St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana